Water Purification

(WT Bot – Autonomous Water Testing Robot)

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Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology Specialized in Information Technology

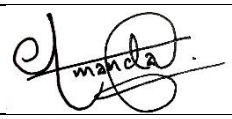
Department of Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT (Maximum 300 words)

Water pollution, driven by industrial discharge, agricultural runoff, and urbanization, poses a critical threat to ecosystem health, food security, and human well-being, particularly in water-scarce regions like Sri Lanka. Effective water resource management hinges on continuous and widespread water quality monitoring. However, conventional methods relying on manual sample collection and laboratory analysis are inherently slow, labor-intensive, expensive, and provide only a sparse, point-in-time snapshot of water body health, failing to capture dynamic pollution events.

This research proposes the development of **WT Bot (Water Testing Bot)**, an autonomous, unmanned surface vehicle (USV) designed to revolutionize water quality monitoring. The WT Bot is a floating robotic platform capable of autonomously navigating diverse water bodies such as rivers, reservoirs, and lakes. It will be equipped with a suite of calibrated sensors to measure critical water quality parameters including pH, turbidity, temperature, and dissolved oxygen in real-time as it traverses its environment.

The system will integrate GPS for geospatial tagging of data, obstacle detection for safe navigation, and wireless communication modules (e.g., 4G/LoRa) to transmit time-stamped, geo-located water quality data to a cloud-based monitoring platform. This platform will visualize the data on interactive maps, generate alerts for anomalies, and provide a historical database for analysis. By automating the entire process from navigation to data delivery, the WT Bot aims to dramatically increase the efficiency, coverage, and frequency of water quality monitoring, enabling proactive environmental management and early warning of contamination events.

Keywords: Water Purification, Machine Learning, Artificial Intelligence, Internet of Things (IoT), Water Quality Monitoring, Alkaline Water Generation, Distilled Water System, Autonomous Water Testing Robot, Autonomous Surface Vehicle (ASV), Unmanned Surface Vehicle (USV), Environmental Robotics, Real-time Sensors, GPS-enabled Data

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Finally, my heartfelt thanks go to my family and friends for their unwavering encouragement, understanding, and patience throughout the demanding phases of this academic pursuit. Their belief in my work has been a constant source of motivation.

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LIST OF ABBREVIATIONS

Abbreviation	Description
WT Bot	Water Testing Bot
AI	Artificial Intelligence
IoT	Internet of Things
GPS	Global Positioning System
pH	Potential of Hydrogen
DO	Dissolved Oxygen
MCU	Microcontroller Unit

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1. INTRODUCTION

1.1 Background and Context

Access to clean water is a fundamental human right and a cornerstone of public health and sustainable development. In Sri Lanka, surface and groundwater bodies are the lifeblood of communities, supporting drinking water supplies, agriculture, and fisheries. However, these vital resources are under increasing stress from anthropogenic activities. Rapid industrialization, intensive agricultural practices employing fertilizers and pesticides, improper waste management, and urbanization have led to the degradation of water quality nationwide [6], [14].

The North Central Province, for instance, faces a severe public health crisis with the prevalence of chronic kidney disease of unknown etiology (CKDu), which has been strongly linked to the consumption of water contaminated by heavy metals and agrochemicals. This highlights a critical failure in our ability to continuously monitor and safeguard our water sources. Current monitoring practices, managed by bodies like the Central Environmental Authority (CEA) and the National Water Supply and Drainage Board (NWSDB), are predominantly manual. Teams are dispatched to collect samples from fixed locations on a periodic basis (e.g., quarterly). These samples are then transported to centralized laboratories for analysis, a process that can take days or even weeks [1], [15].

This reactive approach is fundamentally inadequate. It fails to capture transient pollution events from industrial spills or agricultural runoff that occur between sampling intervals. It cannot provide a high-resolution spatial map of contamination across a large reservoir. It is also resource-intensive and can expose personnel to hazardous environments (e.g., near industrial outfalls or during floods) [1]. There is an urgent, unmet need for a paradigm shift towards **continuous, automated, and wide-area water quality monitoring** to enable proactive management and early warning [7], [10].

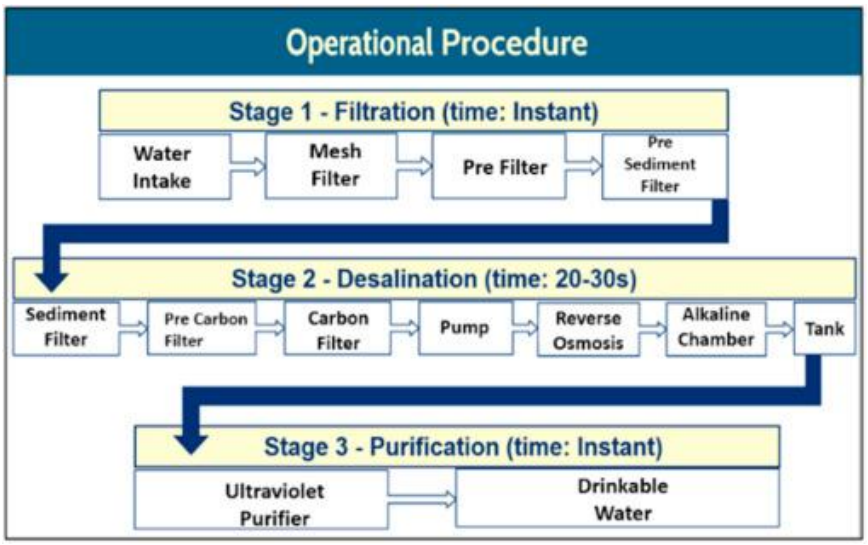


Figure 1. An overview of the water purification process

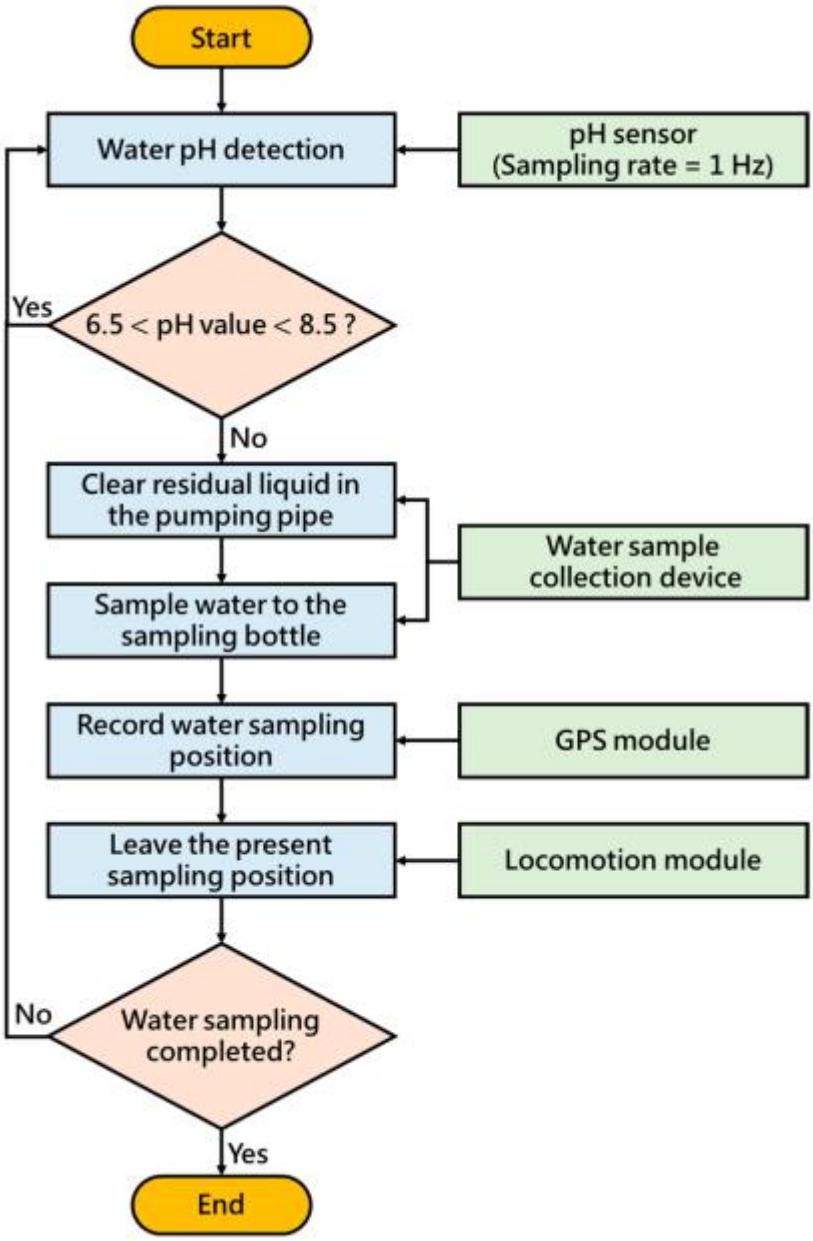


Figure 2. Water quality monitoring process

- This flowchart depicts the operational logic of an automated water sampling system, likely for a platform like the WT Bot, which uses real-time pH data to trigger targeted sample collection. The system continuously monitors pH at 1 Hz; if the reading falls within the normal range (6.5–8.5), it simply clears the sampling line and continues monitoring. However, if an abnormal pH value is detected, indicating a potential contamination event the system automatically initiates a sampling sequence: it collects a water sample into a physical bottle, records the precise GPS coordinates of the anomaly, and then commands the robot to move on. This process repeats until the survey is complete, creating an efficient, event-driven strategy that collects physical evidence only when and where it matters most.

1.2 Literature Review

The field of water quality monitoring has evolved significantly, moving from purely manual methods to incorporating advanced technologies.

- **Traditional Methods:** The gold standard remains grab sampling followed by laboratory analysis (e.g., using spectrometry, chromatography). While accurate, this method is low-frequency, costly, and cannot represent spatial or temporal variability (Chapman, 1996) [1].
- **Static IoT Sensor Networks:** The advent of IoT enabled the deployment of stationary sensor buoys or nodes that transmit real-time data on parameters like temperature, pH, and conductivity (Kumar & Singh, 2021) [4]. Projects like the Smart Finland buoy network demonstrate the power of real-time data. However, the fundamental limitation is their fixed position. A network of buoys provides data only at their specific locations, potentially missing "hotspots" of pollution elsewhere in the water body. They also require significant investment for wide coverage.
- **Remote Sensing:** Satellite and aerial imagery can be used to estimate certain water quality parameters like chlorophyll-a or turbidity over large areas (Giardino et al., 2019) [3]. While excellent for broad-scale synoptic views, this method is limited to surface-level observations, can be obstructed by weather or vegetation, and often lacks the resolution and accuracy for detailed, parameter-specific analysis (e.g., dissolved oxygen).
- **Robotic and Autonomous Systems:** This is the most recent frontier. Research has focused on Autonomous Underwater Vehicles (AUVs) and Unmanned Surface Vehicles (USVs) for environmental monitoring [2], [7].
    - **AUVs** like the ones used in oceanography can carry sensors for deep-water profiling but are often complex and expensive [8].
    - **USVs** offer a more accessible platform for inland waters. Dunbabin & Marques (2012) reviewed various robotic systems for environmental monitoring, highlighting their potential for high-resolution spatial data collection [5]. Rahman et al. (2021) specifically developed a catamaran-style USV for lake monitoring, demonstrating its ability to autonomously follow waypoints and collect basic water quality data. However, many of these research platforms are custom-built, lack integrated AI for anomaly detection, and are not designed for cost-effective deployment in developing nations [7]. The WT Bot project aims to build upon this work by creating a modular, affordable system with integrated data transmission and cloud analytics tailored to the specific challenges of the Sri Lankan context.

Table 1.Water Quality Sensors – Technical Specifications

Technical Parameters

Model		MPG-5199Mini
Measurement configuration		pH/Conductivity/Residual chlorine/Turbidity/Dissolved oxygen/Temp
Range	pH	0-14pH
	Conductivity	0-2000uS/cm
	Residual chlorine	0-20 mg/L
	Dissolved oxygen	0-20 mg/L
	Turbidity	0~20NTU , 0~100NTU
	Temperature	0-60℃
Resolution & Accuracy	pH	Resolution:0.01pH ; Accuracy:±0.1pH
	Conductivity	Resolution:0.1μS/cm ; Accuracy:± 1%FS
	Residual chlorine	Resolution:0.01mg/L ; Accuracy:±2%FS
	Dissolved oxygen	Resolution:0.01mg/L ; Accuracy:±0.3mg/L
	Turbidity	Resolution:0.01NTU; Accuracy:±2% or ±0.02NTU
	Temperature	Resolution:0.1℃ ; Accuracy:±0.5℃
Communication Interface	RS485	
Power Supply	AC 220 V 50W	
Working Temperature	0 ~ 50℃	
Inlet	6mm	
Outlet	16mm	
Protection	IP65	
Wireless	Optional(for Mobile App and Website)	
Housing material	ABS resin waterproof cabinet	
Cabinet size	260mm×190mm×165mm（L×W×H）	

- Detailed specifications of sensors used in WT Bot including measurement range, accuracy, and communication protocol.
- MPG-5199Mini multi-parameter water quality analyzer directly integrate water quality parameters in one cabinet: pH, Conductivity, Residual chlorine, Turbidity, Dissolved oxygen and temperature; all water quality sensors are digital type with RS485, easy to use: plug and use; all operation is easy to on 4-inch color touch screen. The system integrates all functions in one: online water quality analysis, remote data transmission, data recording, alarm, calibration, which provides great convenience for the collection and analysis of various water quality data. Mainly application: Drinking water pipe network, secondary water supply, swimming pool water, direct drinking water, surface water, aquaculture and other normal temperature and pressure environments.

Table 2. Water Quality Parameters and Sensor Specifications

Water quality standards provide by World Health Organization.

Parameters	Units	WHO Standards [25]	Analytical Methods [26]
pH	-	6.5–8.5	pH meter
EC	µs/cm	400	Electrometric
Temp	°C	12–25	Digital thermometer
DO	mg/L	4–6	Winkler method

- Summarizes the target water quality parameters (pH, Turbidity, Temperature, Dissolved Oxygen) with their measurement ranges, resolution, and accuracy requirements based on WHO standards.

Table 3. Function comparison of the proposed multi-function unmanned surface vehicle (MF-USV) with some existing USVs/autonomous surface vehicles (ASVs)

Reference	Function	Sensor	Algorithm
Villa et al. [5]	Obstacle avoidance	LiDAR	LOS-based GNC Algorithm
Wang et al. [8]	Obstacle avoidance	Navigation radar	IACO Algorithm
Kim et al. [9]	Obstacle detection	Vision sensors	Skip-ENet Algorithm
Steccanella et al. [10]	Waterline and obstacle detection	Camera	U-net CNN Algorithm
Guardeño et al. [11]	Obstacle avoidance	LiDAR	ATESOA algorithm
Li et al. [12]	Obstacle avoidance	Microwave radar, 4G Camera	–
	Water quality monitoring	Chl-a turbidity, water dissolved oxygen, water conductivity, oxidation-reduction potential, water temperature, salinity, pH sensors	–
Ferri et al. [6]	Obstacle avoidance	Laser scanner, Sonar, Altimeter	VFH$^+$ algorithm
	Water quality monitoring	Chemical sensors (Hg, Cr, Cd, and dispersed oil)	–
Madeo et al. [13]	Water quality monitoring	pH sensor, ORP sensor, salinity sensor, dissolved oxygen probe	–
Cao et al. [14]	Water quality monitoring	pH sensor, TDS sensor, turbidity sensor	Ensemble learning algorithm

Reference	Function	Sensor	Algorithm
Cryer et al. [15]	Water quality monitoring	CTD, DO, pH, pCO2, nitrate, chlorophyll fluorescence, CDOM, turbidity, DOC sensors	–
Kong et al. [7]	Water surface cleaning	Vision module, Grasper	YOLOv3 algorithm
Wang et al. [23]	Water surface cleaning	Conveyor belt	–
Ruangpayoongsak et al. [24]	Water surface cleaning	Camera, waste scooper	–
Li et al. [25]	Water surface cleaning	Camera, collection box	YOLOv3 algorithm
Proposed MF-USV	Obstacle avoidance	Ultrasonic Sensors	Threshold-based obstacle avoidance algorithm
	Water quality monitoring	pH sensor	–
	Water surface cleaning	Vision sensor, salvage net	Hue-based color filtering algorithm

- Oceans and rivers cover about 71% of the Earth's surface and support billions of aquatic organisms, but human activities have caused severe water pollution through industrial waste, sewage, radioactive materials, and plastic waste. To address these environmental challenges, researchers have developed autonomous systems such as unmanned surface vehicles (USVs), autonomous surface vehicles (ASVs), and autonomous underwater vehicles (AUVs) for applications including navigation, obstacle avoidance, water quality monitoring, and water surface cleaning. These systems use advanced sensors and algorithms to collect real-time environmental data and improve monitoring efficiency. However, many existing systems focus on only a single function. Therefore, this study proposes a multi-function unmanned surface vehicle (MF-USV) capable of performing obstacle avoidance, water quality monitoring, and water surface cleaning simultaneously. The system integrates modules for navigation, positioning, monitoring, communication, and remote control, enabling efficient and real-time water environment monitoring and maintenance.

1.3 Research Gap

While the global landscape of environmental monitoring has seen advancements in remote sensing and automated sensor networks, a significant and critical gap persists in the practical, scalable, and intelligent monitoring of inland water bodies, particularly within the context of developing nations like Sri Lanka [1], [4]. Current methodologies, even when incorporating modern technology, fail to provide a holistic, real-time, and actionable picture of water quality. This project addresses this multifaceted gap, which can be deconstructed into three primary dimensions:

1. The Mobility and Coverage Gap: From Static Points to Dynamic Plumes.

Current monitoring paradigms are trapped between two suboptimal extremes. The first is the traditional, manual approach: deploying field teams to collect grab samples at sparse, fixed locations. This method is inherently slow, labor-intensive, and provides only a fragmented, time-averaged snapshot, entirely missing transient pollution events like industrial discharge or agricultural runoff that can occur and dissipate between sampling cycles [4].

The second extreme is the deployment of fixed sensor stations. While capable of high-frequency data logging, these stations are expensive to procure, install, and maintain, limiting their number. Consequently, they offer data only at a single point, rendering them incapable of mapping the spatial heterogeneity of a water body. They cannot locate the source of a pollutant entering a reservoir or track the full extent and movement of an algal bloom. There is a distinct absence of a **mobile, autonomous, and affordable middle ground** platform that can bridge the gap between sparse point samples and non-existent continuous spatial data, allowing authorities to survey entire water bodies on-demand and with high granularity [2], [5], [10].

2. The Data Integration and Accessibility Gap: From Raw Logs to Actionable Intelligence.

The pathway from a physical water sample to a management decision is often fractured and delayed. In manual methods, the chain of custody, transport to a central lab, and subsequent analysis can take days or weeks. Even with automated systems, the data is frequently logged locally and requires physical retrieval or is transmitted to proprietary, inaccessible databases.

This creates a critical lack of **seamless, real-time data integration with geospatial context**. Decision-makers, such as regional environmental officers, rarely have access to a unified platform where they can intuitively visualize current water quality. The ability to see a live digital map of a reservoir color-coded by turbidity or overlaid with a heatmap of dissolved oxygen, accessible from a standard web browser, represents a transformative leap in situational awareness that is currently unavailable. Without this integration, data remains a static record rather than a dynamic tool for proactive intervention.

3. The Intelligence and Proactivity Gap: From Passive Logging to Adaptive Response.

The most advanced gap lies not in data collection, but in data interpretation and autonomous reaction. Current robotic monitoring efforts, where they exist, largely function as automated data loggers. They follow a pre-programmed path, record measurements, and the data is analyzed only after the mission is complete. This approach misses the opportunity for **in-situ, real-time analysis and adaptive response**.

There is a distinct technological gap in implementing "on-edge" intelligence. By processing data directly on the robot, the system could identify an anomalous reading a sudden pH drop, a spike in conductivity and react immediately. This could trigger a real-time alert to authorities ("Potential pollution event detected at coordinates X, Y") or dynamically alter the robot's mission ("Investigate source of anomaly by performing a high-resolution grid search in this area"). This shift from passive logging to an **intelligent, adaptive sensing system** is crucial for capturing ephemeral events and providing the earliest possible warning, a capability entirely absent from current operational frameworks [7], [13].

The WT Bot project is uniquely positioned to close this tripartite gap. It proposes a holistic system that directly tackles the lack of mobility with an autonomous surface vehicle, bridges the data accessibility divided with a cloud-based geospatial platform, and lays the foundation for future intelligent monitoring by enabling on-edge anomaly detection and adaptive mission control. This creates not just a new tool, but a new paradigm for practical, proactive, and intelligent water resource management.

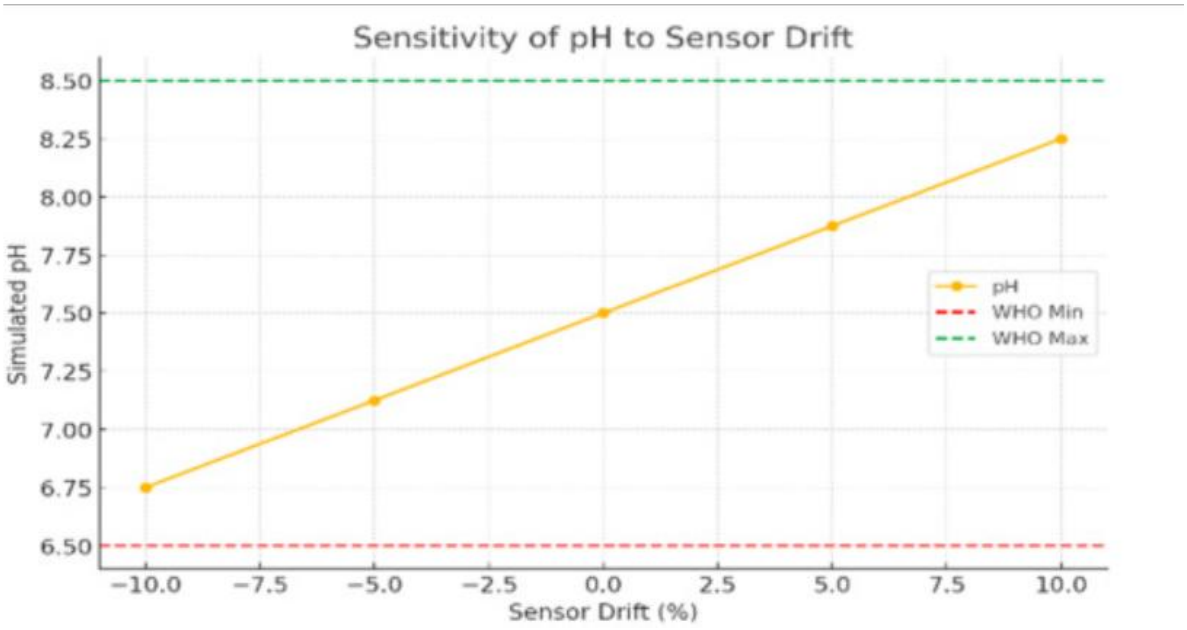


Figure 3. Sensitivity of pH to Sensor Drift

- This image presents a line graph illustrating the sensitivity of simulated pH levels to sensor drift, expressed as a percentage on the x-axis ranging from -10% to +10%. The y-axis shows the corresponding simulated pH values, which span from approximately 6.50 to 8.50. Three distinct lines are plotted: one representing the simulated pH, which increases steadily as sensor drift moves from negative to positive values, and two horizontal dashed lines indicating the WHO minimum and maximum recommended pH thresholds. The graph demonstrates that as sensor drift deviates from zero in either direction, the simulated pH shifts accordingly, with positive drift leading to pH values above the WHO maximum and negative drift causing readings below the WHO minimum, highlighting the critical impact of sensor accuracy on water quality assessment.

2. OBJECTIVE

2.1 Main Objective

To design, develop, and validate an **Autonomous Water Testing Robot (WT Bot)** capable of performing real-time, geo-referenced water quality monitoring across inland water bodies, thereby providing a more efficient, safer, and data-rich alternative to manual sampling methods.

2.2 Specific Objectives

1. **Robotic Platform Design and Fabrication:** To design and construct a stable, durable, and maneuverable floating catamaran-style platform (or similar hull design) capable of housing all electronic components, sensors, and power systems, while operating reliably in outdoor freshwater environments.
2. **Sensor Integration and Data Acquisition:** To integrate a suite of water quality sensors (pH, Turbidity, Temperature, Dissolved Oxygen) with a central microcontroller and develop the necessary firmware to collect, timestamp, and pre-process sensor data accurately and reliably.
3. **Autonomous Navigation and Control System:** To implement an autonomous navigation system using a GPS module for waypoint following and electronic speed controllers (ESCs) for precision motor control. This includes integrating ultrasonic or infrared sensors for basic obstacle detection and collision avoidance.
4. **Real-Time Data Telemetry and Cloud Platform:** To establish a robust wireless communication link (using Wi-Fi, 4G, or LoRa) to transmit geo-tagged water quality data from the robot to a cloud-based server (e.g., AWS, Firebase) in real-time.
5. **Data Visualization and User Interface:** To develop a web-based or mobile dashboard that visualizes incoming data on a map interface, displays time-series graphs for each parameter, generates alerts for user-defined thresholds, and allows data export and historical analysis.
6. **System Validation and Performance Evaluation:** To conduct rigorous testing of the complete WT Bot system in a controlled environment (e.g., university pool/tank) followed by field trials in a real-world location (e.g., a local lake or stream) to evaluate its navigation accuracy, sensor reliability, data transmission integrity, and overall operational endurance.

3. METHODOLOGY

The development of the WT Bot will follow an iterative; engineering design process divided into distinct phases:

- **Phase 1: Component Selection and Platform Design**
    - **Hull Design:** A catamaran-style hull will be designed using CAD software (e.g., Fusion 360) to provide stability and payload capacity. It will be fabricated from lightweight, waterproof materials like PVC or ABS plastic.
    - **Propulsion:** Two brushless DC motors with propellers and waterproof electronic speed controllers (ESCs) will be used for differential thrust, enabling turning and station-keeping.
    - **Core Electronics:**
        - **Microcontroller:** An Arduino Mega or a Raspberry Pi (depending on processing needs for future AI integration) will serve as the main controller. The Raspberry Pi offers more power for potential on-edge image processing or ML models.
        - **Sensors:** A sensor shield or custom PCB will connect the MCU to the sensor probes: a DS18B20 for temperature, a analog turbidity sensor, a gravity analog pH meter, and a DO probe (e.g., from Atlas Scientific).
        - **Navigation:** A high-sensitivity GPS module (e.g., NEO-6M) will provide positional data. Ultrasonic sensors (HC-SR04) will be mounted on the front for obstacle detection [5], [10].
        - **Power:** A high-capacity rechargeable Li-Po battery (e.g., 12V, 10Ah) will power the system, managed by a power distribution board.
        - **Communication:** A 4G/LTE shield (for cellular networks) or a long-range LoRa module will be used for data transmission, depending on the target deployment area's connectivity [4].
- **Phase 2: Firmware and Software Development**
    - **Navigation Logic:** The microcontroller will be programmed (C++ for Arduino, Python for Raspberry Pi) to execute a navigation routine. This involves reading GPS coordinates, calculating bearing to the next waypoint, and adjusting motor speeds accordingly. A simple obstacle avoidance routine will pause forward motion if an object is detected within a critical distance.
    - **Data Acquisition and Logging:** The firmware will continuously read sensor values at a defined frequency. Each reading will be time-stamped and tagged

with the current GPS coordinates. Data will be logged locally on an SD card as a backup and prepared for transmission.

- **Communication Protocol:** The system will establish a connection to the cloud platform and publish data packets in a structured format (e.g., JSON) via MQTT or HTTP protocols.

- **Phase 3: Cloud Platform and Dashboard Development**
  - **Backend:** A cloud database (e.g., Firebase Realtime Database or InfluxDB) will be set up to store incoming time-series and geo-spatial data.
  - **Frontend Dashboard:** A web application will be developed (using React, Node-RED, or similar) to fetch data from the database. It will feature an interactive map (Leaflet/Mapbox) to plot the robot's path and display sensor readings at each point. It will also include time-series charts (using Chart.js) for each parameter and a configurable alert system.
- **Phase 4: Integration, Testing, and Validation**
  - **Component Testing:** Each subsystem (navigation, sensing, communication) will be tested individually.
  - **Pool Testing:** The fully assembled robot will be tested in a controlled pool to verify waterproofing, basic maneuvering, sensor response, and data transmission.
  - **Field Trial:** A final field trial will be conducted in a local lake or reservoir. The robot will be programmed to navigate a set of waypoints covering different areas. The data collected will be compared against manual grab samples analyzed with a calibrated benchtop meter to validate the robot's sensor accuracy. System endurance, communication reliability, and navigation precision will be documented.

Table 4. Parameters of sensors mounted on the MF-USV

Sensor Type	Image	Range	Working Voltage (V)	Accuracy	Producer
GPS		≥−161 dBm	3.3~5.0	Position: 2.5 m Velocity: 0.1 m/s Heading: 0.5°	U-BLOX [1]
Ultrasonic		2~400 cm	3.3~5.0	0.3 cm	ELECFREAKS [2]
pH		0~14	5.0	±0.1 pH	DFROBOT [3]
Vision		Lens field of view: ±37.5° horizontal ±23.5° vertical	6.0~10.0	Resolution: 1280 × 800 dpi	PIXYCAM [4]

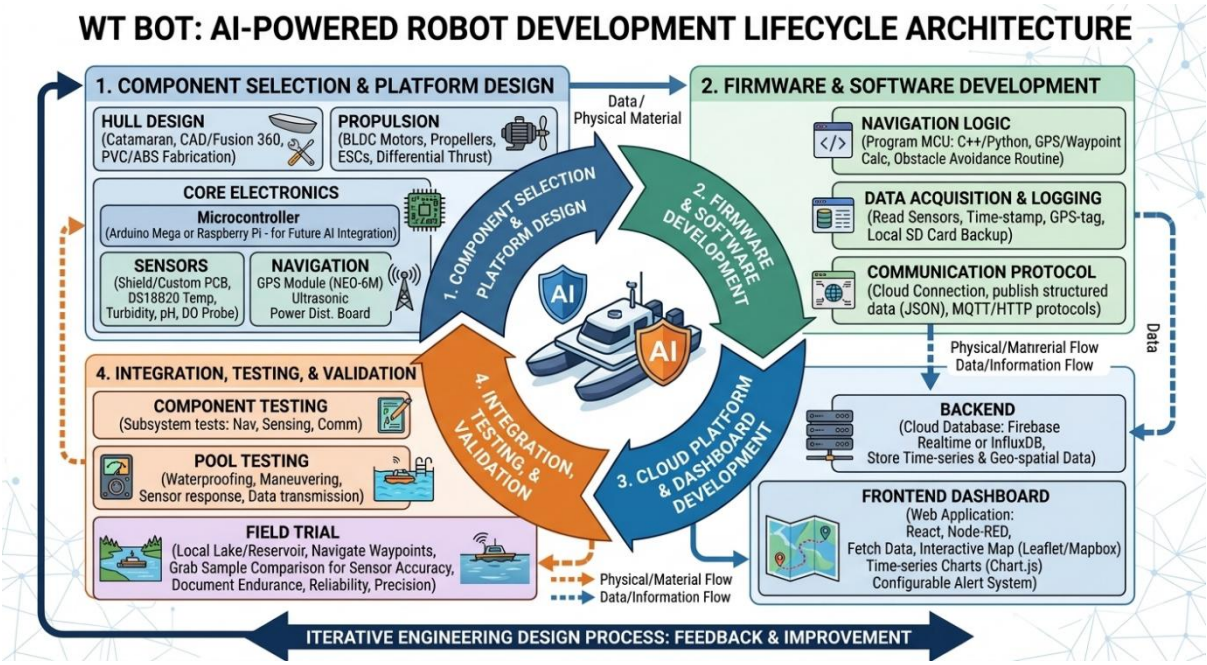


Figure 4. WT BOT: AI-Powered Robot Development Lifecycle Architecture

4. HIGH-LEVEL SYSTEM ARCHITECTURE

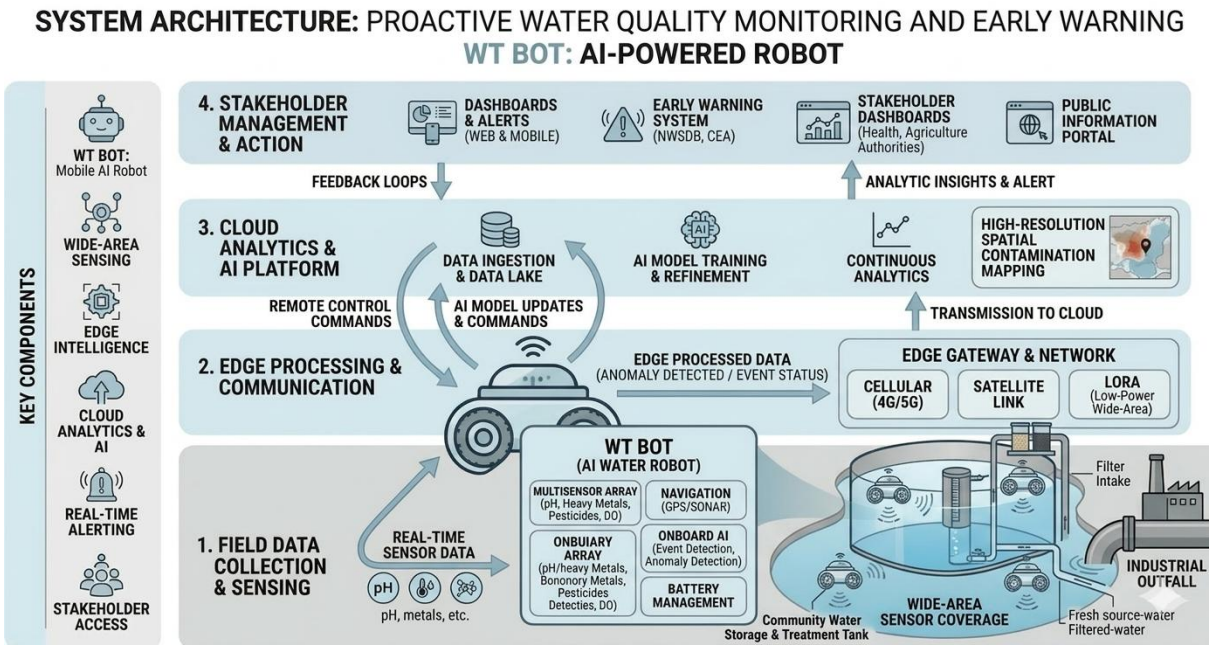


Figure 5. High-Level System Architecture

5. USER REQUIREMENTS

Functional Requirements

- The WT Bot shall measure water quality parameters including **pH (range 0-14), Turbidity (0-1000+ NTU), Temperature (0-50°C), and Dissolved Oxygen (0-20 mg/L)**.
- The robot autonomously navigates to a series of pre-programmed GPS waypoints with an accuracy of ±3 meters.
- The robot shall be able to avoid obstacles (e.g., boats, large branches) by detecting them with proximity sensors and altering its coutheir or stopping.
- The system shall transmit geo-tagged water quality data packets to a cloud server at a minimum frequency of once every 10 seconds.
- The robot shall log all data locally on an SD card as a backup in case of communication failure.
- The user dashboard shall display the robot's real-time position on a map and allow users to view historical paths.
- The dashboard shall generate time-series graphs for each water quality parameter.
- The system shall allow users to define threshold values for parameters (e.g., pH < 6.0) and generate a visual/email alert when exceeded.

Non-Functional Requirements

- **Reliability -** The robot's hull and electronics enclosure must be waterproof (IP64 or higher rating) to withstand splashes, rain, and potential minor leaks during operation.
- **Endurance -** The system should be capable of continuous operation for a minimum of **2-3 hours** on a single battery charge to allow for meaningful survey missions.
- **Accuracy -** Sensor readings should be within ±5% of the readings from a calibrated professional-grade water quality meter.
- **Usability -** The user dashboard must be intuitive and require minimal training to operate, with clear data visualization and simple mission controls.
- **Maintainability -** The system design should be modular, allowing for easy replacement of sensors, propellers, or batteries in the field.
- **Safety -** The robot must have a manual override function via a radio controller (RC) for emergency retrieval.

6. COMMERCIALIZATION PLAN

The WT Bot presents a viable product with multiple pathways to market, addressing a clear need for environmental monitoring.

- **Target Market Segments:**
    1. **Government Agencies:** The primary market includes the Central Environmental Authority (CEA), National Water Supply and Drainage Board (NWSDB), and Mahaweli Authority of Sri Lanka. These bodies are mandated to monitor water quality and would benefit from increased efficiency and data coverage.
    2. **Research and Academia:** Universities and research institutions (e.g., NARA - National Aquatic Resources Research and Development Agency) conducting limnology or environmental impact studies could use the WT Bot for high-resolution data collection.
    3. **Private Sector:** Industries with environmental compliance obligations (e.g., tea factories, textile manufacturers, hotels with effluent treatment plants) could use the WT Bot for self-monitoring of downstream water bodies.
    4. **NGOs and International Aid Organizations:** Groups like UNDP, World Bank, or local NGOs working on water resource management projects could deploy the WT Bot as a tool for community-based monitoring and project assessment [9], [15].
- **Business Model:**
    - **Product Sales:** Direct sale of the complete WT Bot system, including the robot, base station, and software license.
    - **Service Model:** Offering "Water Quality Monitoring as a Service," where the company operates the robot and provides reports and data access to clients, removing the need for them to invest in and maintain the technology.
    - **Data Subscription:** For larger networks, a subscription model could provide access to the cloud platform, data storage, and advanced analytics.
- **Competitive Advantage:** The key differentiators are its **autonomy, affordability, and integration with a user-friendly cloud dashboard**. Compared to importing expensive, specialized research USVs, a locally developed WT Bot can be offered at a fraction of the cost, with local support and customization for Sri Lankan water bodies [6], [14].

- **Future Development Roadmap:**
    - **V2.0:** Integration of a robotic arm for physical sample collection. Addition of a multi-parameter sonde for more advanced parameters (nitrates, phosphates, heavy metals).
    - **V3.0:** Implementation of AI-based "smart navigation" where the robot identifies potential pollution plumes (e.g., from a thermal or turbidity gradient) and autonomously decides to investigate them further.

7. BUDGET AND JUSTIFICATION.

Table 5.Budget and Justification

Item	Component/Description	Estimated Cost (LKR)	Justification
1. Core Electronics	Microcontroller (Raspberry Pi 4/Arduino Mega), Power Distribution Board, Cables, Connectors	1,000	The central processing unit: a Pi offers more scope for advanced features.
2. Water Quality Sensors	pH Electrode & Circuit, Turbidity Sensor, DS18B20 Temp Sensor, Dissolved Oxygen Probe	3,500	These are the primary data collection instruments. Quality sensors are critical for accuracy. DO probe is the most expensive.
3. Power System	12V Li-Po Battery (High Capacity), Battery Charger, Voltage Regulators	5,500	Provides operational endurance. A quality Li-Po battery is required for longer missions.
4. Hardware components	SSD	4,500	Covers recurring costs for cloud services and miscellaneous hardware needs.
TOTAL ESTIMATED BUDGET		**14,500**	

8. WORK BREAKDOWN STRUCTURE (WBS)



Figure 6. Work Breakdown Structure

9. GANTT CHART

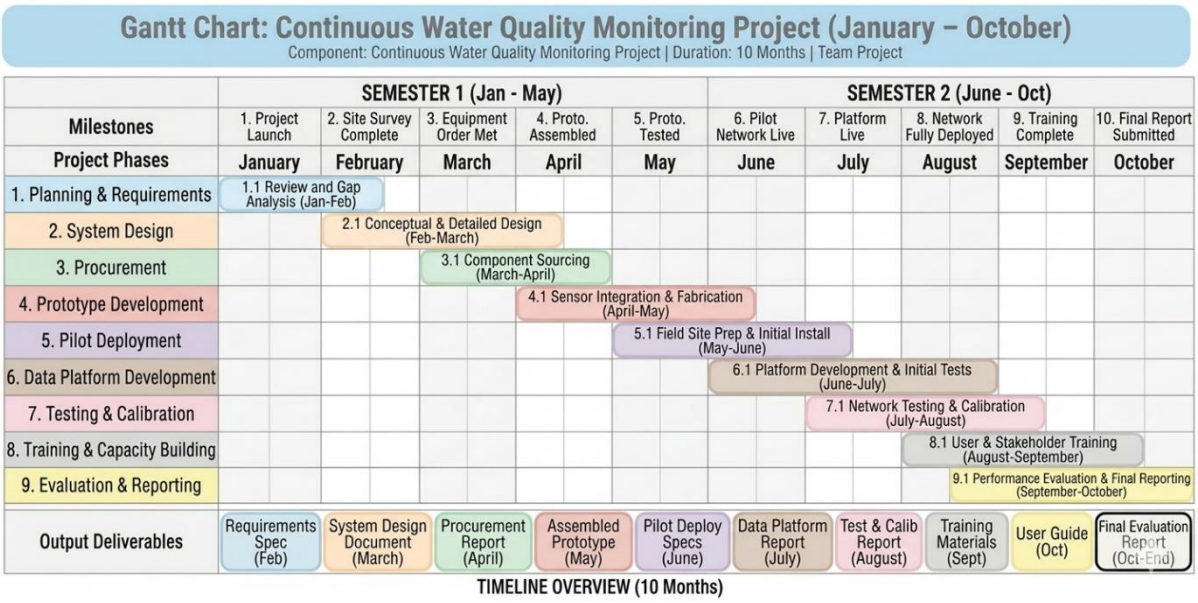


Figure 7. Gantt Chart

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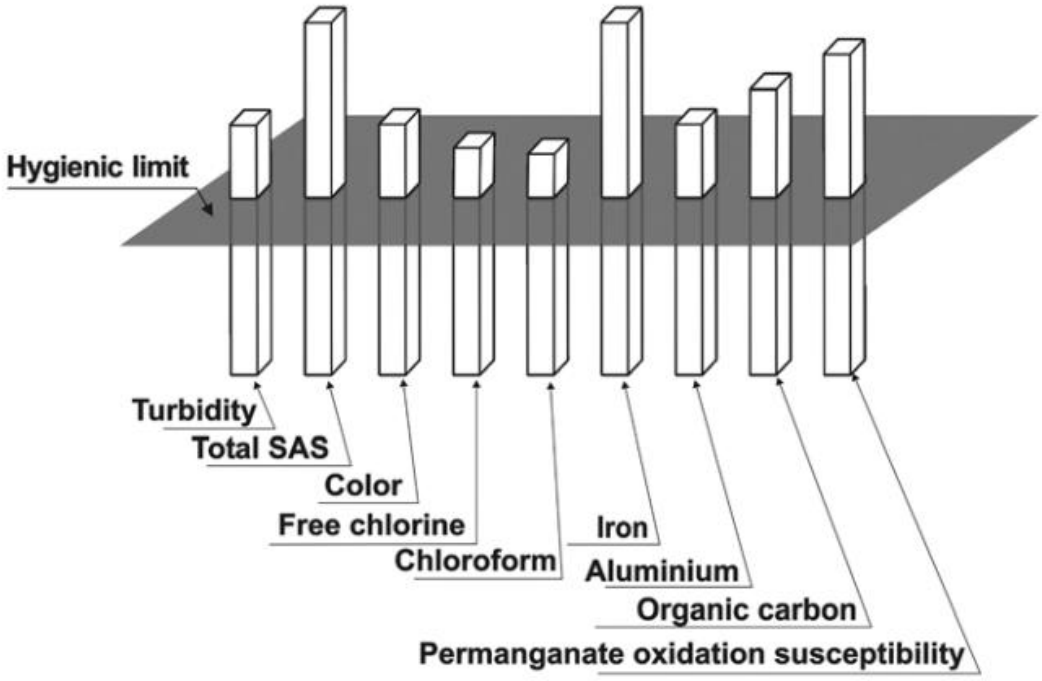
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11.APPENDICES



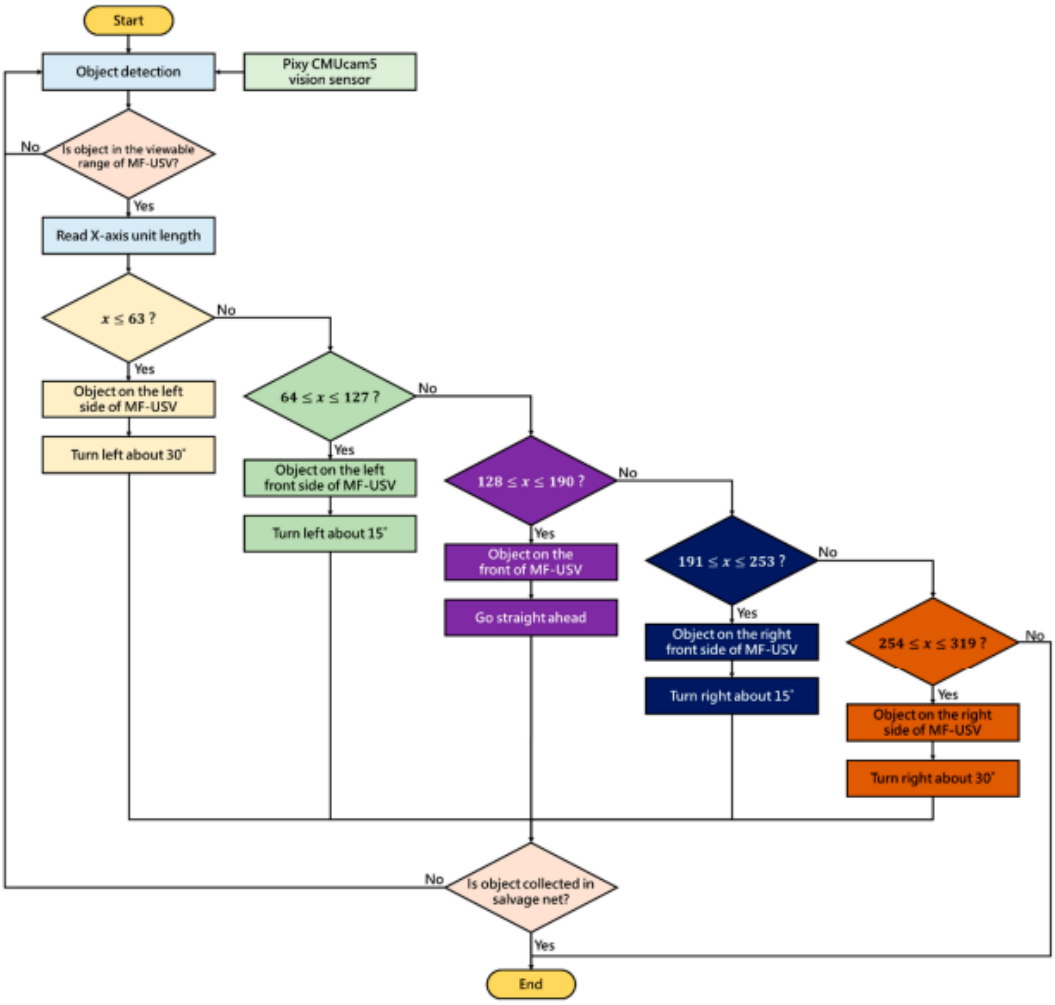
Appendix A. The deviations of tap water quality from hygienic requirements

- The figure illustrates a comparison of several **water quality parameters** against a defined **hygienic limit**, indicating whether each parameter exceeds acceptable standards for safe water use. The parameters shown include turbidity, total surface-active substances (SAS), color, free chlorine, chloroform, iron, aluminum, organic carbon, and permanganate oxidation susceptibility. Each vertical bar represents the measured level of a specific parameter, while the horizontal shaded plane represents the hygienic threshold. Bars extending above this limit indicate contamination levels that exceed recommended safety standards, suggesting potential risks to water quality and public health. This visualization helps identify which contaminants require treatment or monitoring to ensure that the water meets acceptable hygiene and safety regulations.

Parameter	Initial water	Treated water	Standard
Colouration, degree	20 – 50	< 20	20
Odour, points	2 – 3	1	2
Turbidity, mg/L	4 – 14	0.87 – 3	1.5
pH	7.25 ± 0.3	7.32 ± 0.1	6.0 ÷ 9.0
Taste and aftertaste, points	2 – 3	1	2
Residual chlorine, mg/L	1.2	< 0.6	0.8÷1.2
Chloroform, mg/L	0.1	Unfound	0.06
Tetrachloride, mg/L	0.004 – 0.006	Unfound	0.005
Aluminium, mg/L	0.24±0.02	0.11 ± 0.001	0.5
Iron, mg/L	3.0 – 14.0	2.0 – 11.0	0.3
Cadmium, mg/L	0.0005	Unfound	0.001
Strontium, mg/L	0.103	0.09	2.0
Ammonia, mg/L	< 0.04	Unfound	2.0
SAS, mg/L	3.0	<0.025	1.0
SAS, mg/L*	0.5	<0.04	0
Arsenic, mg/L	0.005±0.0001	Unfound	0.05
TMA (per 1 mL) *	63–630	27 – 150	100
Coli-index*	100000	50000	< 3

Appendix B. The efficiency of water treatment

- The table presents a comparison of **water quality parameters for initial (untreated) water and treated water against standard permissible limits**. It shows that several contaminants such as turbidity, aluminum, iron, chloroform, ammonia, arsenic, and cadmium are present in the initial water but are significantly reduced or become undetectable after treatment. Key indicators such as colour, odour, taste, and turbidity also decrease to levels closer to or within the recommended standards. The results demonstrate the effectiveness of the water treatment process in improving water quality by lowering contaminant concentrations and ensuring the treated water meets or approaches acceptable safety standards for human use.



Appendix C. Water surface cleaning algorithm

- The flowchart outlines the decision-making process for an object detection and retrieval system using a Pixy CMUcam5 vision sensor mounted on a Marine-Floating Unmanned Surface Vehicle (MF-USV). It begins by checking if an object is within the sensor's viewable range; if not, it continues searching, but if detected, the system reads the object's X-axis coordinate to determine its position relative to the vehicle. Based on the coordinate ranges, the object is classified as being on the left, left-front, front, right-front, or right side, triggering corresponding actions such as turning left or right by 15° or 30°, or going straight ahead. After maneuvering, the system checks whether the object has been collected in the salvage net, repeating the process until successful collection and termination.

Algorithm 2 The WSCPP algorithm (for irregular region).

Input: Irregular region with coordinates of boundary points in the region $S' = \{N_1, N_2, \ldots, N_m\}$ width of single effective cleaning d, obstacle set $\{A_1, A_2, \ldots, A_r\}$
Output: Optimal trajectory L_{best}

$L_{best} = \emptyset$
Combine the points in S' with points in $\{A_1, A_2, \ldots, A_r\}$ as point set P
Apply triangulation to P and get triangle set [35]. Delete the triangles that are contained in the obstacles, and get the final triangle set T
Combine triangles in set T into multiple convex polygons, and get the sub-region set $B = \{C_1, C_2, \ldots, C_s\}$
Choose the sub-region that contains original boundary points and closest to the start point of the robot as the first sub-region C_f, and $C_{tmp} = C_f$.
while $B! = \emptyset$ **do**
    Delete C_{tmp} from B
    Generate coverage planning path L_{tmp} using Algorithm 1.
    if $L_{best}! = \emptyset$ **then**
        Define line x as the line connecting the end point of L_{best} to the first point of L_{tmp}.
        Get all the obstacles that have intersection points with x, and sort the obstacles from the shortest to the longest according to their distances to the end point of L_{best}, and obstacles set is $O = \{A'_1, A'_2, \ldots, A'_k\}$.
        Define the path bypassing the obstacles as u
        for $i = 1$ *to* k **do**
            Calculate the intersection points p_1, p_2 of x and A'_i
            Divide the boundary of A'_i into two parts m_1, m_2 by points p_1, p_2
            if both m_1 and m_2 are inside of the region S' **then**
                Add the shorter one of m_1 and m_2 into u
            else
                Add one of m_1 and m_2 that is inside of the region into u
            end if
            Add u into L_{best}
        end for
    end if
    Add trajectory point set L_{tmp} into L_{best}
    if there are sub-regions adjacent C_{tmp} in B **then**
        Set C_{tmp} as the sub-region adjacent and closest to the end point of L_{tmp}.
    else
        Set C_{tmp} as the sub-region closest to the end point of L_{best}.
    end if
end while

Appendix D. WSCPP algorithm - Model Training Code

- The WSCPP algorithm plans an optimal cleaning path for a robot in an irregular area with obstacles. It first divides the region into convex sub-regions by triangulating boundary and obstacle points. Starting from the sub-region nearest the robot, it generates coverage paths and connects them by navigating around obstacles using the shortest viable boundary segments, ultimately building a complete and efficient trajectory.